Ogbonna Arinze

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SUMMARY

Frontend Engineer with 8 years of experience architecting and owning modern front-end ecosystems, from developer-facing dashboards and design systems to real-time, data-intensive applications. Deep expertise in React, TypeScript, and Next.js, with a track record of building reusable component libraries, establishing comprehensive testing protocols (unit, integration, E2E), and implementing CI/CD pipelines and client-side performance monitoring. Comfortable navigating early-stage codebases and shaping product architecture from the ground up, with hands-on experience in real-time data streaming (SSE, EventSource) and full-stack AI product development.

CORE COMPETENCIES

- **Frontend Architecture:** Scalable component systems, reusable design systems (Radix UI), monorepos (Nx, Turborepo), SSR/SSG, and responsive, accessible UI development.
- **Languages & Frameworks:** TypeScript, modern JavaScript (ESNext), React, Next.js, Vue.js, NodeJS, Redux, Zustand, and Tailwind CSS.
- **Real-Time & Streaming:** Server-Sent Events (SSE), EventSource API, real-time data visualization (Recharts, react-chartjs), and WebSocket patterns.
- **Testing & Quality:** Playwright (E2E), Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven automated QA.
- **Performance & Observability:** Lighthouse, Chrome DevTools, React Profiler, code splitting, lazy loading, CDN strategies, Sentry, Datadog RUM, and bundle analysis.
- **CI/CD & Infrastructure:** GitHub Actions, Docker, AWS (CloudFront, S3), Vercel, automated deployment pipelines, and Nx monorepo orchestration.
- **Design & UX:** Figma-to-code workflows, custom design systems, accessibility (WCAG/ARIA), and UI component documentation with Storybook.
- **Leadership:** Technical architecture ownership, mentorship, code reviews, Agile/Scrum, and product-minded engineering.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, owning the frontend architecture and optimizing workflows for clinical site integrations.
- Architected a **reusable design system** and internal component library in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and **Chrome DevTools** profiling.
- Co-developed a full-stack **Next.js** real-time chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to query their data via a conversational interface.
- Implemented **JWT-based authentication** and **role-based access control** on the frontend using **TypeScript** and **React Query**, with HTTP-only cookies for secure token handling and permission-based UI rendering.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring **Slack notifications** for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality through structured code reviews.

Descasio Lagos, Nigeria (Hybrid)
Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, improving speed and user experience.
- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports.

- Utilized **Storybook** for documenting React components, streamlining change management and ensuring clear communication of component APIs and design patterns.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React, Redux, RxJS**, and **Axios** for asynchronous data handling.
- Adopted the native **EventSource API** for **real-time data streaming**, replacing socket.io for faster updates and a more responsive interface with lower latency.
- Built dynamic **data visualization dashboards** using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**, enabling real-time transaction monitoring.
- Developed a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Transformed user journeys and **Figma** designs into reusable **React** components, ensuring a consistent, intuitive user interface aligned with design specifications.
- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas.
- Engineered a **CI/CD pipeline** for automated deployment of APIs to testing environments on **Heroku** and production servers on **AWS**.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, and devised comprehensive documentation detailing application functions and capabilities.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app with 22 interactive chart types (**Recharts**), admin dashboard, document ingestion pipeline, marketing landing page, and a Remotion video generation service.
- Built the **Next.js** frontend with **React 19**, **Tailwind CSS v4**, and **Recharts**, featuring responsive mobile layouts, dark mode, URL state management with nuqs, and **SSE streaming** for real-time chat responses with 15s keepalive heartbeats.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, and memory limits.
- Integrated **Langfuse** for LLM observability, **Infisical** for secrets management, and **Traefik** for reverse proxy routing, with SEO optimization (Open Graph, JSON-LD, SSR).

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*